Decision Science - Supplier Management for Olist

Data-Driven Decision Support for Profit Optimization

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Executive Summary

Objective: Help Olist CEO evaluate seller-level profitability.

- Dataset includes seller sales, reviews, and profit-impacting metrics.
- Analysis indicates that 29% of sellers generate negative value.
- To get the highest profits, after removing 855 bad suppliers, Olist's Profit would increase 403,367 BRL, Profit Margin would increase 1%
- Recommendation: implement a dynamic performance-based seller removal strategy.

Business Context

What is Olist? Brazil's top B2B platform connecting sellers to online marketplaces.

Revenue Model:

- 10% commission on product price
- 80 BRL monthly subscription per seller

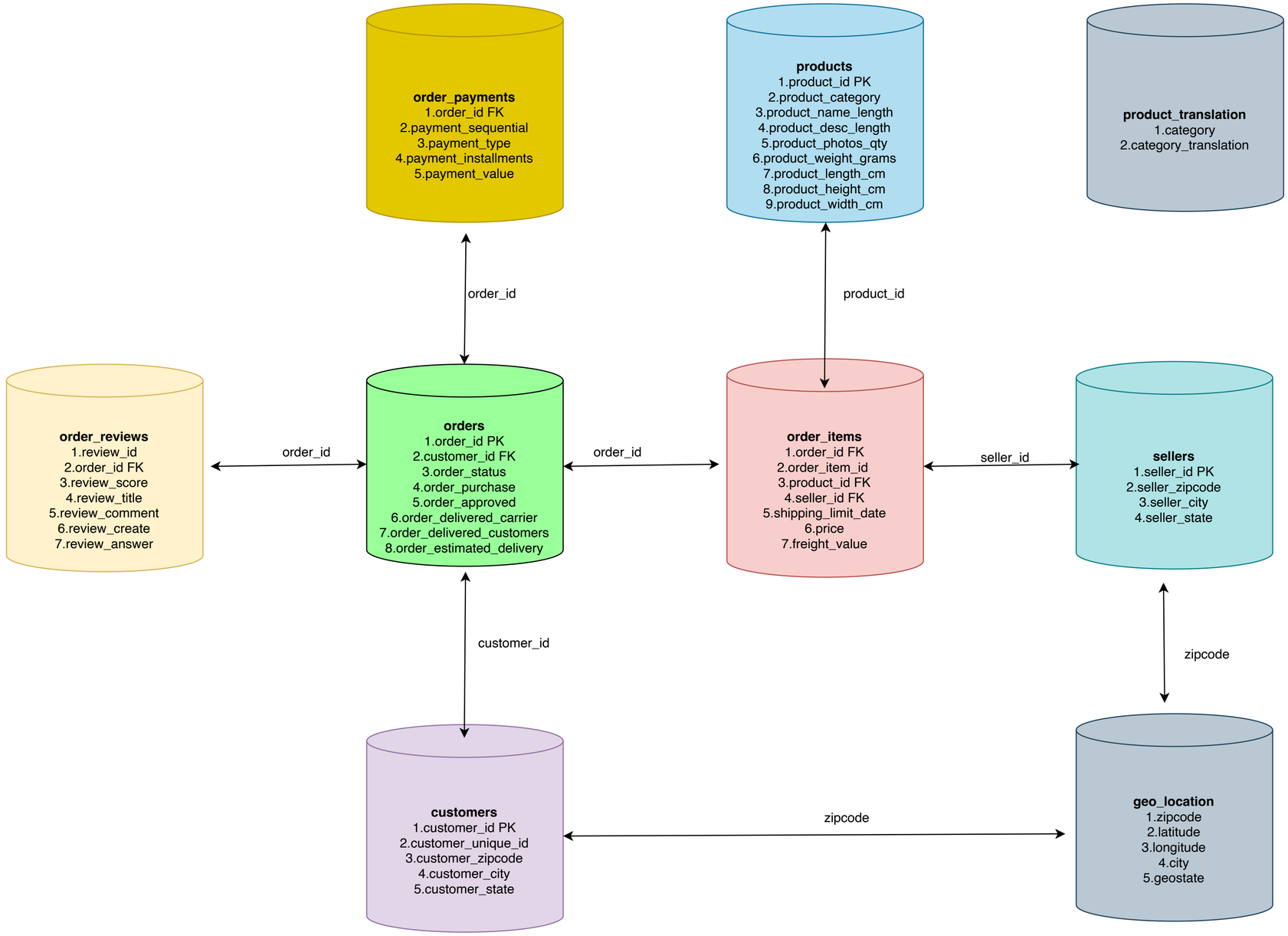
Cost Drivers:

- Bad review impact (support, refund, reputation)

  An assumption about the monetary cost:
  '1 star': 100, '2 stars': 50, '3 stars': 40, '4 stars': 0, '5 stars': 0

- IT cost scales with sqrt(#sellers and #items): scale with the square root of the total number of sellers that have ever joined the platform, as well as with the square root of the total cumulated number of items that were ever sold.

DataSet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6.product_weight_grams
7.product_length_cm
8.product_height_cm
9.product_width_cm
product_translation
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3.review_score
4.review_title
5.review_comment
6.review_create
7.review_answer
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6.order_delivered_carrier
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zipcode
geo_location
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2.latitude
3.longitude
4.city
5.geostate

Financial Overview

Revenues: BRL 2.79M (Sales fees: BRL 1.36M, Subscriptions: BRL 1.43M)

Costs: BRL 2.12M (Review: BRL 1.62M, IT: BRL 0.5M)

Net Profit: BRL ~667k

Why removing the bad suppliers will increase our Profits?

1. Bad suppliers will cost us more Reputation fees

Consumers prone to give lower review score for the bad suppliers, which would cost Olist more money to fix the reputation and win back the market. It would cost Olist avarage 393 BRL more to fix the reputation for the bad suppliers.

	review_score	cost_of_reviews	revenues	profits
mean	4.101513	545.402764	938.934427	393.531663
min	1.000000	0.000000	0.650000	-21519.517000
max	5.000000	39400.000000	24387.263000	18067.605000

Why removing the bad suppliers will increase our Profits?

2. Less items/ less suppliers we use would save us the IT cost fee

Using 2967 supplier would cost Olist around 500K BRL, while removing 855 bad suppliers, it saves around 104K BRL in the IT Cost fees

- Current IT Cost is 500K BRL
- After removing bad suppliers the IT cost is around 396K BRL

What-if Analysis: Profit Optimization

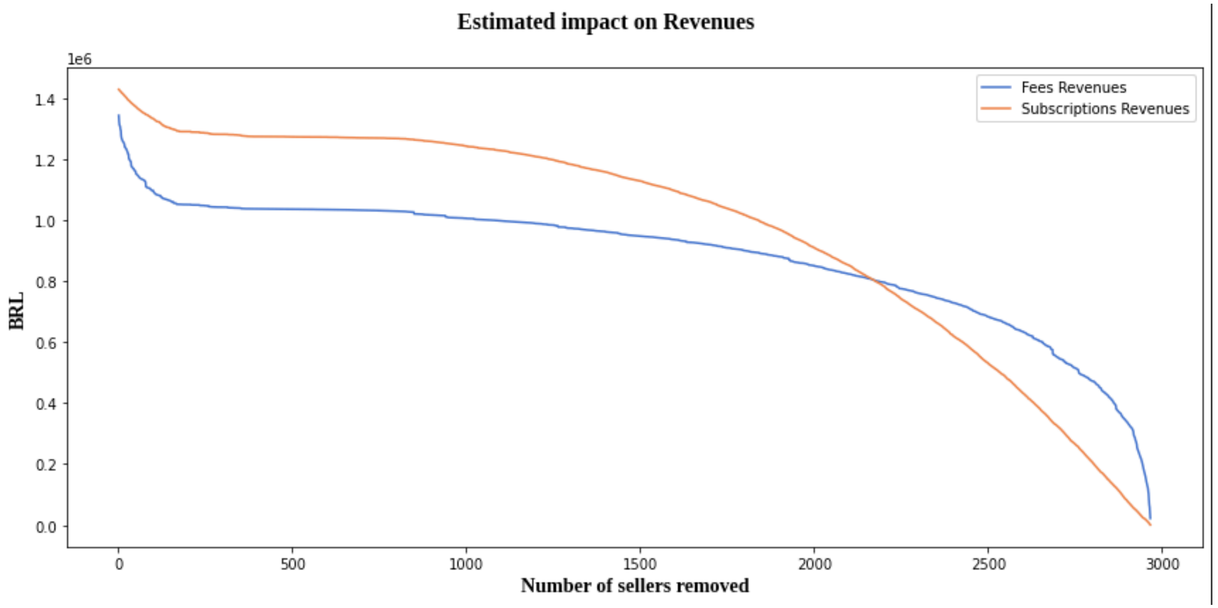
Need to find the maximum profits by balancing the cost and revenue

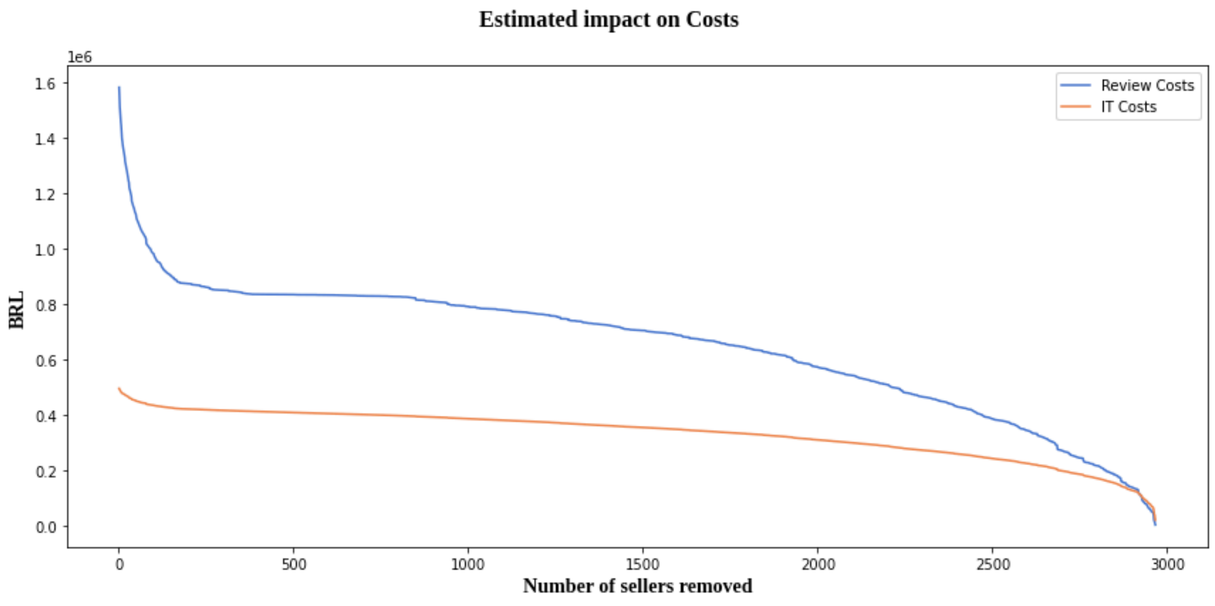
Impact on Revenue

- Subscriptions and fees decline when sellers are removed
- Revenue loss is moderate until ~2000 sellers are removed

Impact on Costs

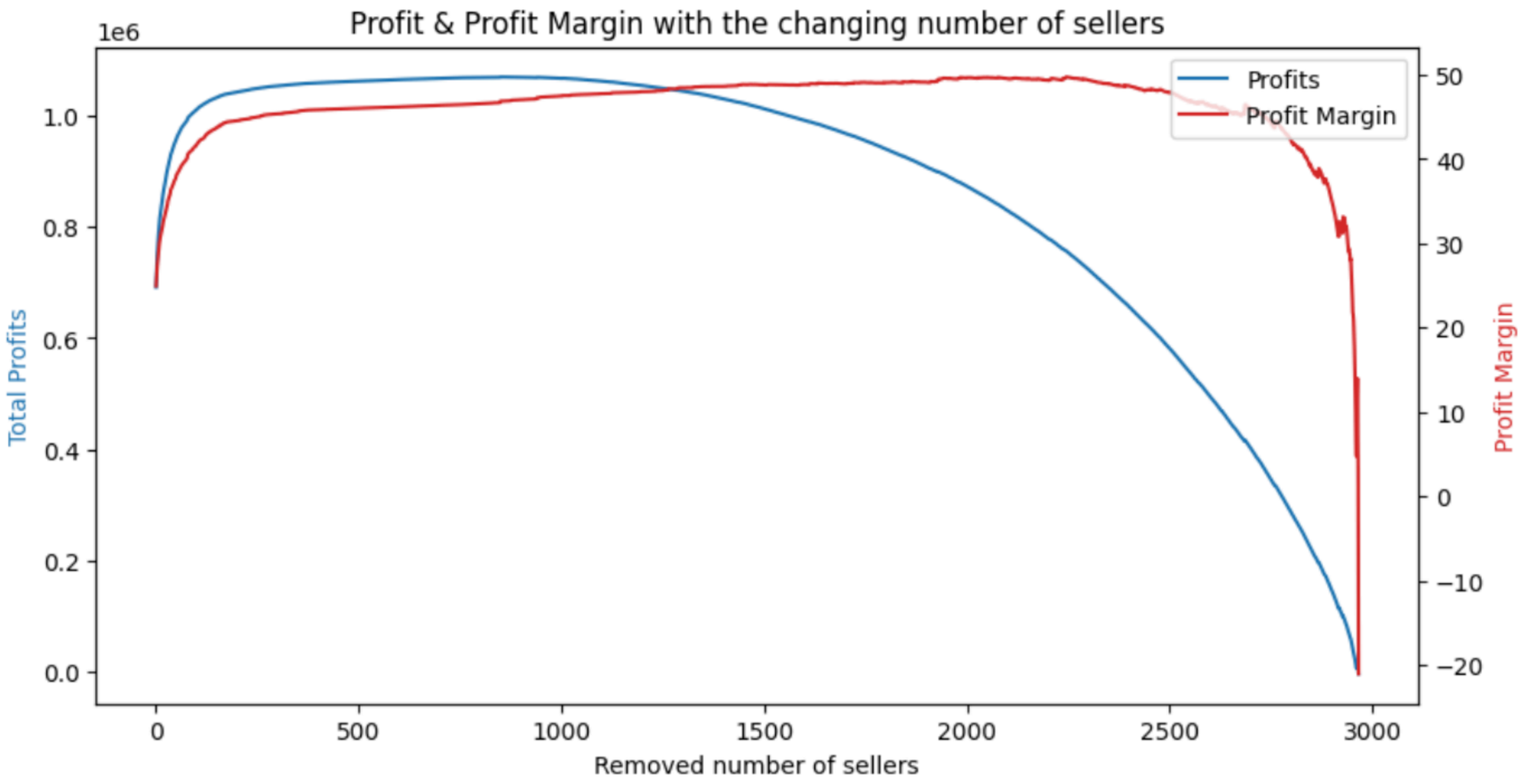
- Review costs and IT costs drop with fewer sellers
- Cost reduction is substantial from early removals





Result

- Max net profit gain when removing 854 sellers (29%)
- Profit margin peaks when removing 2244 sellers (76%)



Appendix

- IT Cost Formula: $IT_cost = \alpha \cdot \sqrt{n_sellers} + \beta \cdot \sqrt{n_items}$
  - Alpha = 3157.27, Beta = 978.23
- Key Datasets:
  - olist_sellers_dataset.csv: seller IDs, registration, activity span
  - olist_orders_dataset.csv: order volumes and timestamps
  - olist_order_reviews_dataset.csv: review scores and comments
- Tools Used:
  - Programming: Python (Pandas, NumPy)
  - Visualization: Plotly, Seaborn, Matplotlib
  - Modeling & Simulation: Custom profit models, what-if simulation, nonlinear scaling logic
- **Environment**: Jupyter Notebook / Google Colab
- **Limitations & Notes**: Revenue estimates exclude logistics cost due to data absence; assumes consistent monthly subscription fees